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Hyperspectroscopic screening of melanoma on acral volar skin

Takashi Nagaoka[1], Atsushi Nakamura[1], Haruka Okutani[2], Yoshio Kiyohara[3], Hiroshi Koga[4], Toshiaki Saida[4] and Takayuki Sota[2]

[1]*Waseda Research Institute for Science and Engineering, Waseda University, Shinjuku, Tokyo, Japan,* [2]*Department of Electrical Engineering and Bioscience, Waseda University, Shinjuku, Tokyo, Japan,* [3]*Dermatology Division, Shizuoka Cancer Center Hospital, Nagaizumi, Shizuoka, Japan and* [4]*Department of Dermatology, Shinshu University School of Medicine, Matsumoto, Nagano, Japan*

Background/purpose: Early detection and proper excision of the primary lesions of melanoma are crucial for reducing melanoma-related deaths. To support the early detection of melanoma, automated melanoma-screening systems have been extensively studied and developed. In this article, a previously reported hyperspectral imager and melanoma discrimination index are applied to the discrimination of acral lentiginous melanoma (ALM) from acral nevus (AN), and their diagnostic performance is reported.
Methods: The index expresses the disordered nature of each lesion including variegation in color based on variation in spectral information obtained from each lesion. Performance of the index has been studied in thirteen cases of ALM and seven cases of AN, obtained from patients and volunteers, all of whom were Japanese.
Results: The index discriminated ALM from AN with a sensitivity of 92% and a specificity of 86%. The area under the receiver operating characteristic curve was 0.97.
Conclusion: The performance of the proposed objective melanoma discrimination index at a molecular pigmentary level approached that of clinical experts, using the three-step algorithm as the gold standard.

Key words: computer-assisted diagnosis – diagnostic imaging – early detection of cancer – melanoma – spectrum analysis – acral lentiginous melanoma

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MELANOMAS ARE seen most frequently on acral skin in nonwhite people. In Japanese people, about half of all cutaneous melanomas affect acral skin, and nearly 30% of these acral melanomas are detected on the sole of the foot; most melanomas that affect acral skin are of the acral lentiginous type (1). In general, the prognosis of acral lentiginous melanoma (ALM) is poor because most are found at advanced stages. Detection of this type of melanoma in early, curable stages is essential to improve the prognosis. On acral skin, however, benign melanocytic nevi are also frequently encountered. Melanocytic nevi are reported to be present on the soles of about 7% of the Japanese population (2). Clinical differentiation between early melanomas and melanocytic nevi on acral volar skin is important.

It has been revealed that ALM can be discriminated with a high degree of accuracy by the accurate identification of a dermoscopic pattern that is peculiar to ALM (3–7). A parallel ridge pattern with prominent pigmentation of the ridges of the skin markings is a characteristic dermoscopic feature commonly detected not only in invasive melanoma but also in melanomas *in situ* affecting acral volar skin. On the other hand, a parallel furrow pattern with pigmentation along the sulci of the skin markings is a characteristic dermoscopic feature detected in the majority of benign melanocytic nevi on acral volar skin. The sensitivity and specificity of the parallel ridge pattern for ALM have been reported to be 86% and 99%, respectively (8). Moreover, automatic discernment of Saida's pattern classification has been reported by Iyatomi and colleagues (9). They report the ability to classify each pattern with a probability of not less than 80%.

In our previous article (10), a melanoma-screening system was proposed. The system accumulates hyperspectral data (HSD) (11, 12) from pigmented skin lesions (PSLs) within a short time period and provides a melanoma discrimination index that is derived from the HSD. Performance of the system has been studied for a small set of PSLs involving the non-glabrous skin of Japanese subjects, which revealed substantial sensitivity and specificity.

In this article, we report the performance of this system in discriminating melanocytic lesions on acral volar skin of Japanese subjects. A total of 20 melanocytic lesions on acral volar skin including 13 melanomas were studied. Sensitivity of 92% and specificity of 86% were achieved. These measures of performance are close to those of the dermoscopic parallel ridge pattern (3–8), indicating that the system is of value also in detecting melanoma on acral volar skin.

Patients and Method

Patients

This study was approved by the Institutional Review Board of Shizuoka Cancer Center (SCC). Written informed consent was obtained before measurements were made. Twenty patients who had consulted the SCC were registered for this research from July 2008 to April 2011. To evaluate the performance of our system reliably, patients who had inflammation and exudates were excluded. Only ALM and acral nevus (AN) affecting volar skin were included in this research. Pigmented lesions of the nail were also excluded. All subjects were Japanese.

Thirteen PSLs were clinically suspected to be malignant, and histological evaluation of biopsy specimens confirmed the diagnosis of ALM. The remaining seven PSLs were AN. Digital dermoscopic images were also obtained for the 20 lesions. Features of these PSLs are summarized in Table 1. Hyperspectral images that were out of focus, were blurred due to severe saccadic movement, or were technically deficient were excluded from the following analysis.

When a PSL was too large to be measured in one scan, the PSL was measured by dividing it into several areas. In this case, several peripheral areas were concentrated on, in which the essential features of melanoma are usually detected, and all HSD were measured at a position where the ratio of healthy skin to PSL was 1:1.

Methods

A hyperspectral imager for PSLs (MSI-02, Mitaka Kohki Co., Ltd., Tokyo, Japan) was used for this study. A photograph of the apparatus and a schematic optical diagram are shown in Fig 1(a) and 1(b), respectively. For details of the system, see reference (9). Briefly, the specifications of this system are as follows: Spectral resolution: 1.2 nm, spectral range: 380–780 nm, measurement area: 6 mm $\times$ 6 mm, spatial resolution: 13 μm, measurement time: about 20 s, power of light source: 150 W. The melanoma discrimination index was derived from HSD as described in detail in reference (9). Note that an averaged spectrum obtained from the sole of a typical healthy Japanese volunteer was used as a reference spectrum. The index measures the spatial variegation in the spectra obtained from a PSL; in other words, it measures irregularity, disorder, and variation in morphology.

Statistical evaluation

Statistically significant differences between the melanoma and non-melanoma groups were analyzed using the Mann–Whitney *U*-test. A *P*-value less than 0.05 was considered statistically significant. Because the number of samples was small, the jackknife method was used to minimize bias, instead of cross-validation. In this approach, one sample is excluded from all samples and discriminant analysis is performed, and the excluded sample is discriminated with the use of the discriminant coefficient so obtained. The same procedure was executed for all samples, and sensitivity and the specificity were calculated from the results of the discriminant analysis. Receiver operating characteristic (ROC) analysis on resubstitution (13) was also performed for the purposes of future study.

Results

Two-dimensional spectral angle-color maps are shown in Fig. 2 for typical cases of both (a) an ALM (case No. 7) and (b) an AN (case No. 15). The probability of finding a given spectral angle for each is shown in Fig. 2(c) by solid black and dotted lines. The corresponding indices have been calculated to be 5.05 and 3.89. Figure 2(c) reveals that the spectral angles are more widely distributed for ALM than for AN. The index is thus larger for ALM than for AN. It should be

TABLE 1. Lesions analyzed in this study

Case no.	Sex/age	Diagnosis	Size of lesion (mm)	Tumor thickness (mm)	TNM classification	Site of lesion	Index
1	M/56	ALM	18 × 15	6.0	pT4bN3M0	Right hallux	5.74
2	M/87	ALM	20 × 15	6.0	pT4bN2bM0	Right hallux	4.61
3	M/62	ALM	32 × 18	5.0	pT4bN1aM0	Right hallux	4.88
4	F/61	ALM	30 × 35	4.0	pT4bN0M0	Right heel	4.71
5	F/59	ALM	45 × 40	5.0	pT4bN0M0	Right heel	5.66
6	F/72	ALM	25 × 45	2.5	pT2bN0M0	Right hallux	5.17
7	F/72	ALM	40 × 35	1.1	pT2aN0M0	Right sole	5.05
8	M/71	ALM	55 × 50	1.1	pT2aN0M0	Right sole	5.44
9	F/43	ALM	15 × 12	0.6	pT1bN0M0	Right hallux	4.73
10	F/66	ALM	1 × 1	0.3	pT1aN0M0	Left sole	4.95
11	M/80	ALM	8.5 × 5.5	0.7	pT1aN0M0	Left sole	5.47
12	F/77	ALM	20 × 16		pT1aN0M0	Left heel	4.85
13	F/66	ALM	6 × 12		pTisN0M0	Right sole	4.41
14	F/45	AN	5 × 4			Left sole	4.55
15	F/35	AN	7 × 2			Left sole	3.89
16	F/53	AN	7 × 5.2			Right sole	3.96
17	F/55	AN	6 × 5			Right heel	4.00
18	F/37	AN	5 × 4			Right heel	4.63
19	F/59	AN	10 × 5			Right sole	4.33
20	F/49	AN	7.5 × 4.1			Right sole	3.95

ALM, acral lentiginous melanoma; AN, acral nevus.

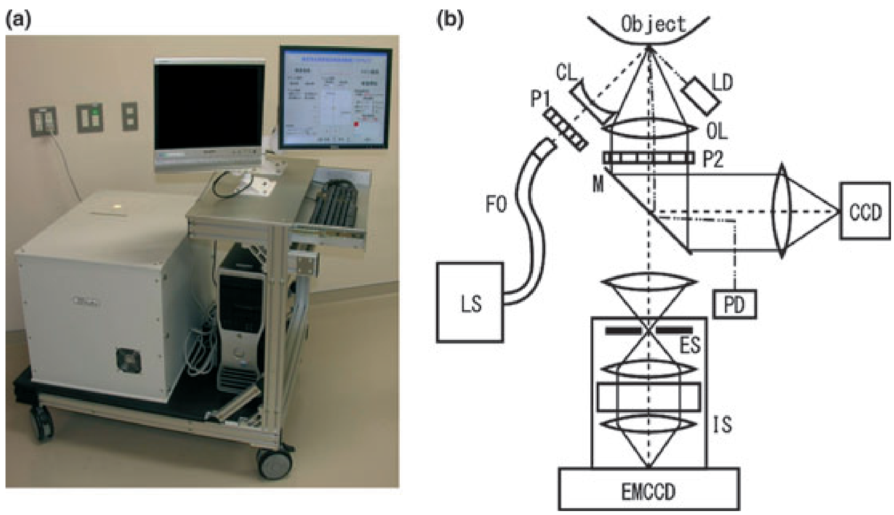

Fig. 1. A hyperspectral imager for pigmented skin lesions. A photograph (a) and schematic optical diagram (b). See reference (9) for details. LS, light source with 150 W halogen lamp; OF, optical fiber; CL, cylindrical lens; P1 and P2, polarizers; M, slanted mirror; CCD, CCD camera monitoring object's image; LD, laser diode; PD, photodiode; IS, imaging spectrograph; ES, entrance slit; EMCCD, electron multiplication CCD camera.

noted that spectral variation doubles when the index increases from 4.0 to 5.0. This is because a logarithmic function is included in the calculation of the index.

There are finite false-positive and false-negative fractions with the use of our index. Two-dimensional spectral angle color maps are shown in Fig. 3 for false-negative cases of (a) an ALM (case No. 13), and for false-positive cases of (b) an AN (case No. 18). The probability of finding a given spectral angle for each case is shown in Fig. 3(c) by solid black and dotted lines. The corresponding indices have been calculated to be 4.41 and 4.63.

Box-and-whisker plots of indices for the two groups are shown in Fig. 4(a). The Mann–Whitney *U*-test reveals that the difference between the medians of indices in the melanoma group and the AN group is statistically significant. Note that index values for each case are shown in Table 1. Figure 4(b) shows the ROC curve on resubstitution. The area under the ROC curve,

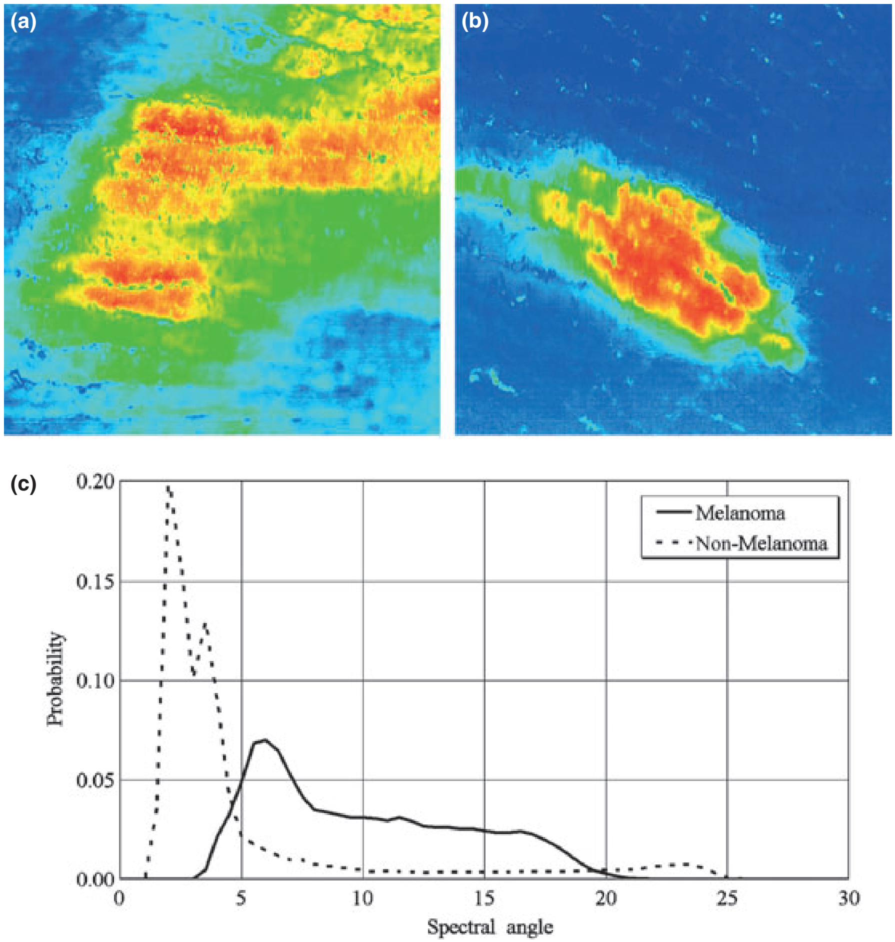

Fig. 2. False-color, two-dimensional angle maps (a), (b) for the acral lentiginous melanoma of case No.7, and the acral lentiginous nevus of case No.15. (c) Respective probabilities of finding a given spectral angle for the acral lentiginous melanoma (Melanoma) of case No. 7, and the acral nevus (non-Melanoma) of case No. 15 are shown in this figure. In the acral nevus, the boundary of the lesion is characterized by a melanin-dominant region, and can be clearly distinguished from normal skin, which is a hemoglobin-dominant region. In contrast, the invasive nature of the melanoma is reflected by nesting of the melanin-dominant region within the hemoglobin-dominant region. The overall features of the probability curve for the acral nevus are obviously different from those for the melanoma. Differences in such overall features of the curve produce the difference in the index proposed in this paper.

AUC, was 0.967, revealing that the index alone is useful for distinguishing ALM from AN. As a result of the discriminant analysis, the optimum threshold value was presumed to be 4.56. A sensitivity of 92.3% and a specificity of 85.7% were obtained by applying this threshold value to the whole dataset. Although sensitivity and specificity do not appear to be particularly high, this translates to incorrect allocation of just one benign and one malignant PSL.

Discussion

It is well known that dermoscopy has introduced a new morphologic dimension in the diagnosis of melanoma. The ABCD rule of dermoscopy (14) summarizes guidelines for diagnosis based on the expression of malignancy or the invasive nature of a melanoma at the morphologic level. Changes in the concentration and distribution of melanin and hemoglobin dominate assessment of the morphology of PSLs. In this study, we attempted to derive a melanoma discrimination index based on data acquired from HSD at a molecular pigmentary level because the index carries information on the spatial changes in concentration of melanin and hemoglobin. Results show that the proposed index is useful in discriminating between ALM and AN. Melanoma cells randomly invade the surrounding normal skin. This random invasion results in the border irregularity that is seen especially at the boundary between the melanoma and the

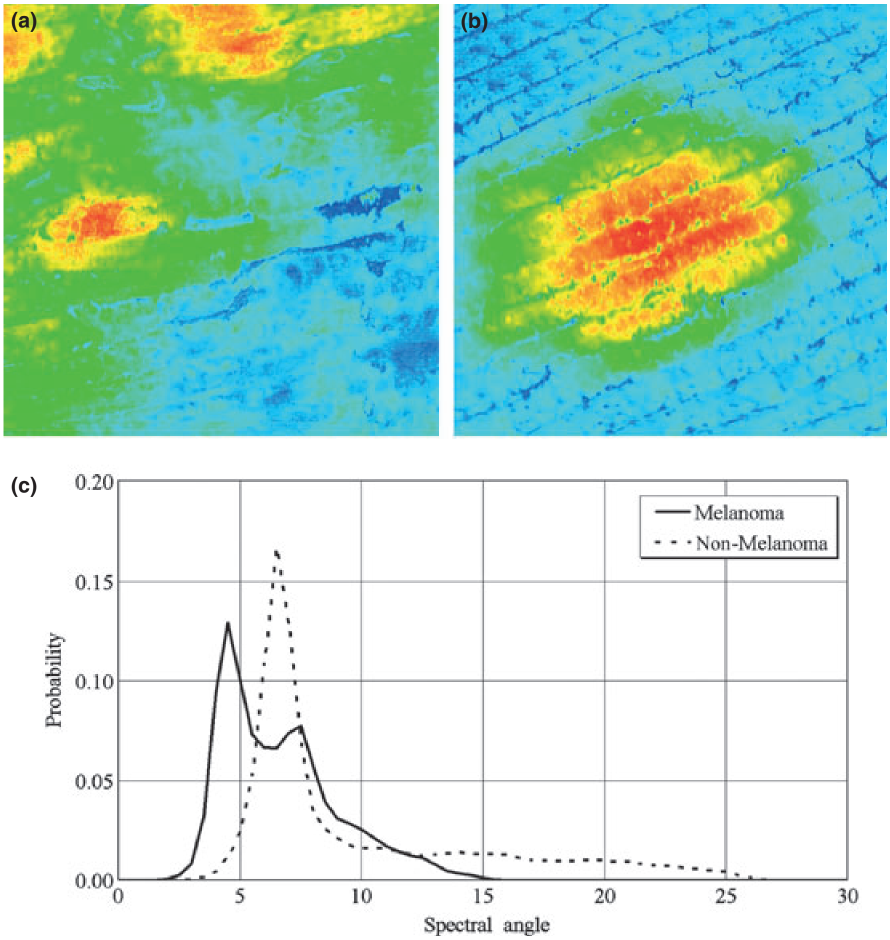

Fig. 3. False-color, two-dimensional angle maps (a), (b) for the acral lentiginous melanoma of case No.13, and the acral lentiginous nevus of case No.18. (c) Respective probabilities of finding a given spectral angle for the acral lentiginous melanoma (Melanoma) of case No. 13, and the acral nevus (non-Melanoma) of case No. 18 are shown in this figure. These two cases are misjudged by our discrimination index. However, because our method of analysis is simple, the cause of the mis-assessment is easily determined.

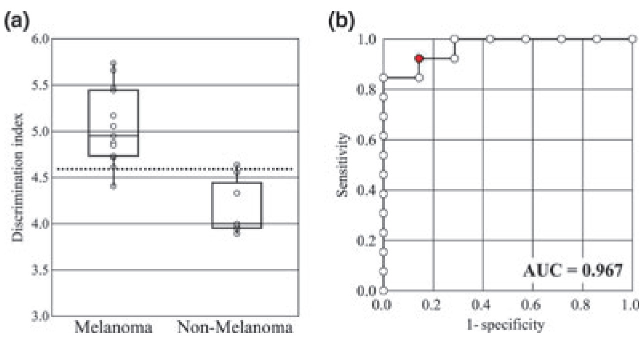

Fig. 4. Box-and-whisker plots of indices for the melanoma and non-melanoma group (a) and ROC curve on resubstitution (b). The AUC attains 0.97. These results show that the index is useful in discriminating melanoma from non-melanoma. A logarithmic function is included in the calculation of our index. Therefore, although the difference between the melanoma group and non-melanoma group appears small, it is actually large.

surrounding normal skin. It results in a wider spectral angle distribution, leading to a rather large index.

We have previously reported automatic discrimination of melanomas on the non-glabrous skin (10). ALM on acral volar skin showed a tendency for the index value to decrease by about 10%, compared with melanomas of the non-glabrous skin. A decrease in the value of our index is attributed to a decrease in the spectral variegation of the PSL. This may be caused by a decrease in the detected light intensity because the thicker stratum corneum of acral volar skin scatters the incident light randomly. It should be noted that the proposed index depends on anatomical location.

As mentioned above, it has been reported that ALM can be discriminated dermoscopically with high accuracy by identifying specific patterns found in acral melanocytic lesions, particularly with the aid of the three-step algorithm (3–8, 15, 16). However, interpretation of dermoscopic features is more or less subjective. The present method objectively and quantitatively discriminates ALM from AN. Although the statistical population is still limited, its diagnostic accuracy appears close to that of the three-step algorithm.

Recently, Swanson et al. (17) have reported that morphologically and physiologically meaningful maps may be derived from 30 spectral radiance images of PSLs, which are recorded at 10 multispectral bands in the 365–1000 nm wavelength region from multiple angles of illumination and detection. They have also shown that the image entropy of these maps may provide a sensitive index that differentiates melanomas from other PSLs. Taking into account the invasive nature of malignant tumors, the analysis of image entropy seems to be a reasonable approach in differentiating between benign and malignant PSLs. We should, however, bear in mind that the reliability of the parameters from which the maps are constructed depends strongly on the underlying model (18, 19).

There are finite false-positive and false-negative fractions in the present analysis. Using cases No. 13 (ALM) and No. 18 (AN), we tried to identify possible reasons why such false discriminations are made. Two-dimensional spectral angle maps for the former and the latter are, respectively, shown in panels (a) and (b) in Fig. 3. The distribution of spectral angles is shown by a bold solid curve for the former and by a dotted curve for the latter in panel (c) of Fig. 3. The present analysis classified case No. 13 (pTisN0M0) as AN because the corresponding index is below the threshold value. The small index value is attributed to the relatively narrow distribution of spectral angles seen in Fig. 3(c). This is a feature found in early-stage melanoma where the production of melanin is still low. Case No. 18 was an AN with a fibrillar pattern, in which filamentous or mesh-like pigmentation made the skin markings appear slanted. Melanin produced in a melanocytic nevus usually appears right above the nests of melanocytes. However, the particular parts of the sole skin receive the physical pressure from the body weight, which induces oblique arrangement of the thick cornified layer. Thereby melanin granules in the cornified layer are also arranged in a slanting manner, which produces the fibrillar pattern (20). Thus, the fibrillar pattern is detected not only in AN but also in ALM, and discrimination of the fibrillar pattern seen in ALM and AM is not always easy. The broader distribution of melanin may make the distribution of spectral angles wider or acquire a longer tail, as seen in Fig. 4(c), leading to a relatively large index value. This is a possible reason why a case with a fibrillar pattern was accidentally classified as an ALM. The new technique of solving this problem will be reported in future publications.

We also wish to draw attention to the following. In calculating the present index, it is not necessary to trace all over the PSL. Many other automatic diagnosis/screening systems for melanoma extract data about the shape of the PSL from the images obtained and subsequently compute a variety of parameters using these data. The extraction of these PSL domain data therefore governs the discrimination accuracy (9).

In conclusion, we propose an objective melanoma discrimination index that works at a molecular pigmentary level and is derived from HSD; we also evaluate its diagnostic performance. This index was highly successful in discriminating between ALM and AN, although the statistical population was small. Despite this limitation, our findings suggest that the disordered nature of each PSL is a very important factor in differentiating ALM from AN we examined using the present system. Further comparison with the diagnoses of experts is needed to confirm the robustness of an index that is based on spatial distribution of spectra. To accumulate such cases, a new, patient- and user-friendly device is being developed. We also plan to increase the number of facilities where the new device is available.

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Address:
Takashi Nagaoka
Waseda Research Institute for Science and Engineering
Waseda University
Shinjuku
Tokyo 1698555
Japan
Tel: +81 55 989 5222
Fax: +81 55 989 6085
e-mail: nagaoka@aoni.waseda.jp